

Free vibrations of a cantilever beam with an added mass

Introduction

Object

Study of a single degree of freedom vibrating system through simulation data and experimental acceleration measurements.

Experimental setup and simulation parameters

The experimental setup consists of a cantilever beam with an added mass on the tip. We have the data from two different accelerometers. The measurements are taken for an Aluminium beam with different parameters.

The beam has rectangular section of thickness 3 mm and height 30 mm. The parameters of the beam are:

$L=45$ cm

Material=Aluminium

Mass on the tip=0.46 kg

The sampling frequency is 2048Hz.

The experiment is composed by 3 tests: the second one is taken as reference using an Aluminium Beam (3*30mm) with a lumped iron mass $m_2=0.46$ kg at the length $L_2=45$ cm; in the others the parameters that have been changed are the lumped mass $m_1=0.2$ kg, for the test one, and the free length $L_3=37$ cm, for the test three.

Purpose

The purposes of this report are:

1. To find natural frequency and modal damping ratio for the first mode as estimated by using the logarithmic decrement method, for experimental measurements.
2. A spectral analysis (FFT) of the free response, showing the first natural frequencies with a graph.
3. A brief discussion about the influence of material, length of the beam and added mass on the first natural frequency.
4. Brief research about approximate formulas available on the web for the Euler-Bernoulli beam theory with added masses (possibly compared to the continuum EB solution without any added mass, for a given length)

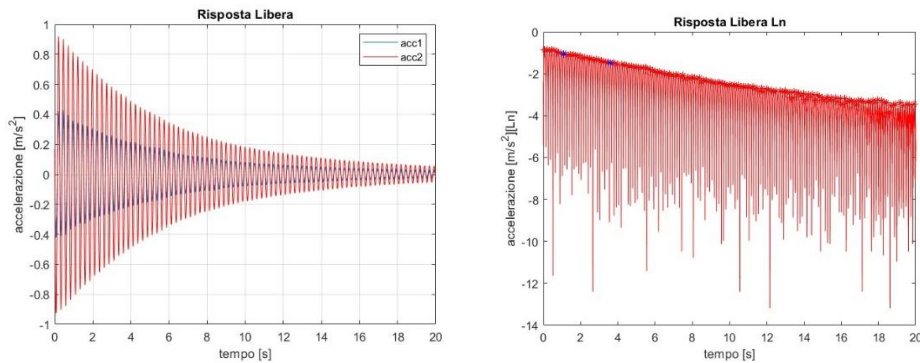
Report

Natural frequency and modal damping

To find the natural frequency, we need to evaluate the period of the signal. To do that, we evaluate the period using the minimum of the $\log|\ddot{x}|$ graph. Then we will read the values of the fifth and the twenty-fifth peaks of the acceleration on the $\log|\ddot{x}|$ graph, to evaluate ζ using the logarithmic decrement method. As ζ known we can compute the damped frequency and the natural frequency of the systems. Changing the parameters of the system we will see that also the natural frequency and the damping ratio will change.

Reference Test

The system taken as reference present the classic behaviour of an underdamped oscillating system. In the figure 1 are shown the acceleration measured by the accelerometers in position $z_1=23$ cm and $z_2=38$ cm. Then we've calculated the logarithm of the modulus of each point to obtain the graph 2.



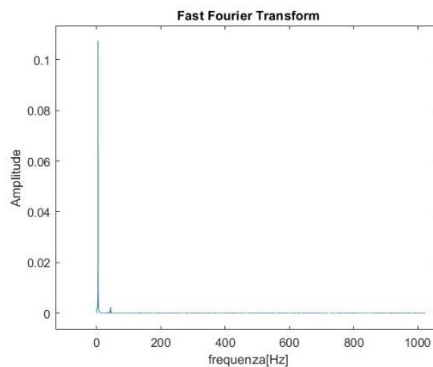
The two points marked in blue on the log graph are the two points used to interpolate the linear decrement, they are far from each other $n=10$ periods so they will be called a_1 and a_{n+1} . Know these two values we can proceed calculating zeta and the damped and the natural frequencies as:

$$\xi = \frac{a_1 - a_{n+1}}{\sqrt{(2 * \pi * n)^2 + (a_1 - a_{n+1})^2}} = 0.0067$$

$$\omega_d = 24.9427 \text{ rad/s}$$

$$\omega_n = 24.9433 \text{ rad/s}$$

The two values are similar due to the small value of zeta (the damping ratio). We can extrapolate the Fast Fourier Transform (FFT) of the free response to see which frequencies are more important of the phenomena.



In this case we can see that the most important frequency is at

$$f = 4 \text{ Hz} \cong \frac{\omega_d}{2 * \pi}$$

In the FFT graph we can see other peaks at different frequencies that are caused by the noise on the measurements.

Results regarding parameter change

Two other experimental tests were carried out by changing the various parameters including the mass and length of the beam. With the first test a mass of 0.2 kg in aluminium was used while for the third test the length of the beam was decreased to 37 cm. At the end of the experimental measurements, it can be said

that in the first test there is a higher natural frequency than in the other two. The following table shows the results of all three experimental measurements:

	TEST1	TEST2	TEST3
ξ	0.0272	0.0067	0.0054
ω_n	35,8272	24.9433	34.269

It is noted that going to change the values of length and mass also change the parameters k and m that govern the trend of the ω_n of the system. By decreasing the mass, the natural frequency of the System will increase, while decreasing the free length of the beam will increase the stiffness of the beam itself and therefore the natural frequency of the system will also increase.

Euler-Bernoulli Model

The Euler–Bernoulli equation describes the relationship between the deflection of the beam and the applied load considering for each point of the beam two degrees of freedom, displacement and rotation:

$$\frac{d^2}{dx^2} \left(EI \frac{d^2 w}{dx^2} \right) = q$$

The displacement $w(x)$ describes the deflection in the x-direction, I is the moment of inertia, E is the modulus of elastic, and q is the distributed load. EI is also called flexural stiffness. By the study of the free body diagram of an infinitesimal part of the beam we reach the D’Alambert wave equation.

$$\frac{\partial^2 w}{\partial t^2} = -\frac{EJ}{\rho A} * \frac{\partial^4 w}{\partial x^4}$$

From this formula, applying the separation of variables, it is possible to obtain the r-th natural frequency of the system.

$$\omega_r = \frac{\pi^2 * r^2}{L^2} * \sqrt{\frac{EJ}{\rho A}}$$

These depend on the boundary conditions of the beam. The beam is clamped in $x = 0$, while in the other extreme it can be found in multiple conditions. In one of these conditions, it could be present an added mass m, the boundary conditions will then be.

$$M(x = L) = 0 \quad , \quad \frac{\partial^2 w}{\partial x^2} = 0$$

$$Q(x = L) = -EJ * \frac{\partial^3 w}{\partial x^3} = -m \frac{\partial^2 w}{\partial x^2}$$

If the mass is not present, then $m = 0$ and the shear force $Q(x = L) = 0$, while the momentum remains null.